Classification Metrics

Classification metrics are essential for evaluating the performance of a classification model. Common metrics include **accuracy, precision, recall, F1 score, and the ROC-AUC**.

These metrics help quantify how well the model predicts the positive class versus the negative class.

Accuracy in Predictions

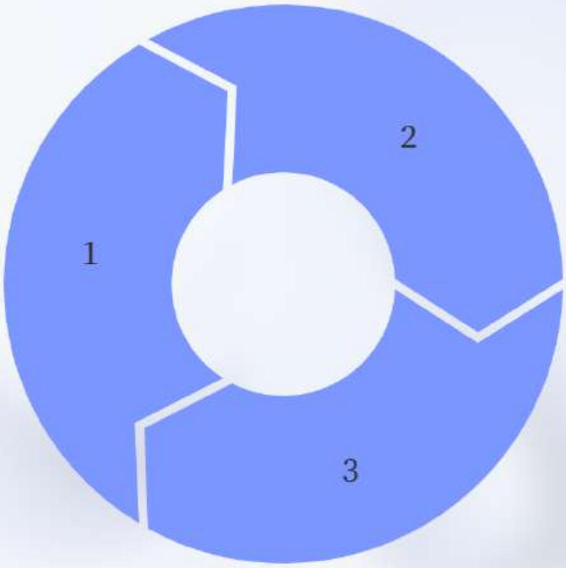
Definition

Ratio of correct predictions to total predictions.

Misleading Nature

Can be misleading for imbalanced datasets.

Understanding Precision



Formula

TP / (TP + FP)

Definition

Measures how many predicted positives are actually positive.

Significance

Important when false positives are costly.

Recall (Sensitivity)

1

Definition

Measures how many actual positives are correctly predicted.

2

Formula

TP / (TP + FN)

3

Importance

Important when missing positives is costly.

4

Applications

Used in medical diagnosis, safety systems.

Understanding F1-Score

Definition

Harmonic **Mean** of precision and recall.

Balance

Balances false positives and false negatives.

Application

Useful for imbalanced datasets.

Performance

Higher value means better model performance.

Understanding ROC-AUC

1 ROC Curve

Plots TPR vs FPR.

2 AUC Metric

Measures separability between classes.

3 Value Range

Ranges from 0.5 (random) to 1 (perfect).

4 Threshold Independence

Performance metric unaffected by thresholds.

Regression Metrics

Mean Absolute Error

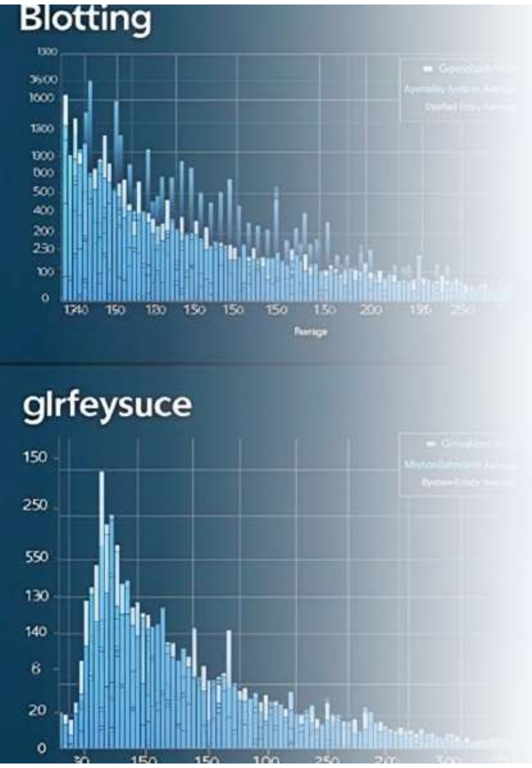
Measures average absolute difference between predicted and actual values.

Mean Squared Error

Calculates the average of the squares of errors - the average squared difference.

R-squared

Indicates the proportion of variance in the dependent variable predictable from the independent variables.



Mean Absolute Error (MAE)

Definition

Average of absolute differences between actual and predicted values.

Interpretation

Easy to interpret (same units as target).

Equal Treatment of Errors

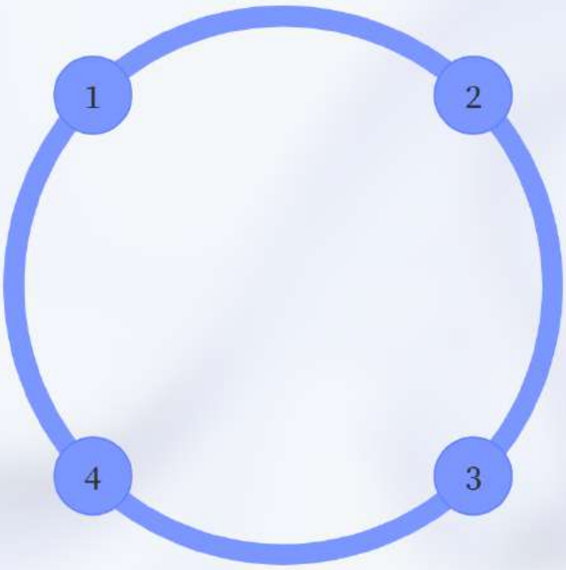
Treats all errors equally.

Outlier Sensitivity

Less sensitive to outliers than MSE.

Mean Squared Error (MSE)

Average of Squared Errors



Penalizes Large Errors

Heavily weights larger prediction errors, highlighting their impact.

Useful for Optimization

Differentiable property aids in optimization problems.

Sensitive to Outliers

Small deviations impact results significantly in the presence of outliers.

Root Mean Squared Error (RMSE)

Definition

Square root of MSE.

Unit Consistency

Same units as target variable.

Error Sensitivity

Highlights large errors strongly.

Usage

Commonly used regression metric.

R^2 Score (Coefficient of Determination)

The R^2 score, also known as the coefficient of determination, is a key statistic that measures how well a model explains the variance in the outcome variable. Its value can range from 0 to 1, and although it can be negative, a higher R^2 score indicates a better fit for the model. However, it may not be a reliable metric when dealing with non-linear models.

Measures Variance

Explains variance in the model.

Value Range

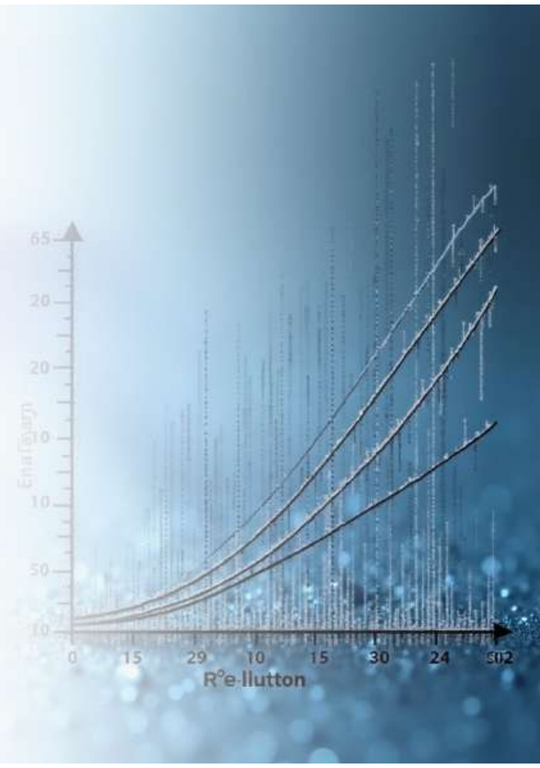
Ranges from 0 to 1.

Higher Value

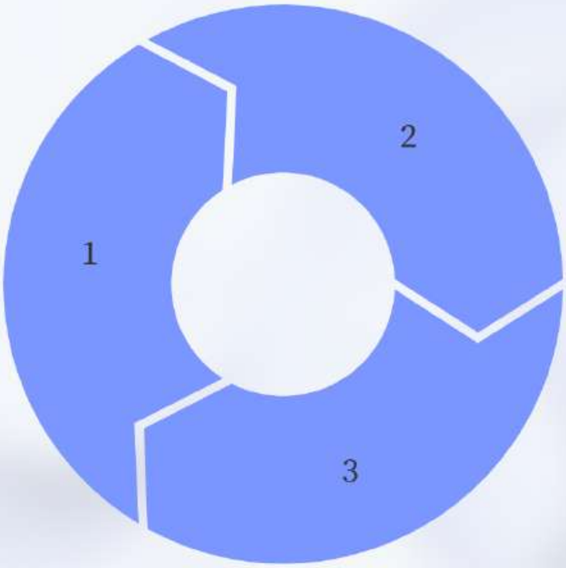
Indicates better fit.

Caution for Non-Linear Models

Not reliable alone.



Unsupervised Learning Metrics



Silhouette Score

Measures how similar an object is to its own cluster compared to other clusters.

Davies-Bouldin Index

Evaluates the average ratio of intra-cluster distances to inter-cluster distances.

Dunn Index

Assesses the ratio of the smallest distance between clusters to the largest intra-cluster distance.

Understanding Silhouette Score

1

Cluster Fit

Measures how well data points fit within clusters.

2

Value Range

Value ranges from -1 to +1.

3

Score Interpretation

Higher score → better-defined clusters.

4

Cohesion and Separation

Considers both cohesion and separation.

Understanding the Davies–Bouldin Index

Average Similarity

Measures average similarity between clusters.

Lower Value Advantage

Lower value indicates better clustering.

Cluster Evaluation

Evaluates cluster compactness and separation.

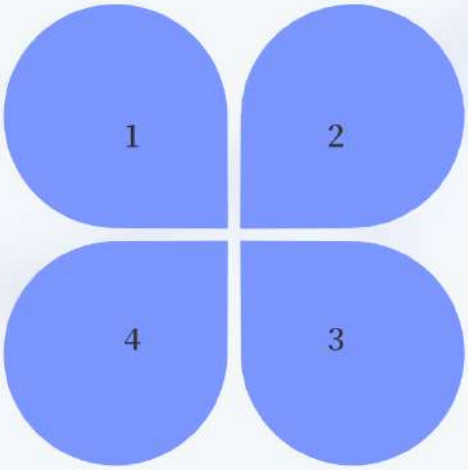
No True Labels Needed

Does not require true labels.

Calinski–Harabasz Index

Variance Ratio

Ratio of between-cluster variance to within-cluster variance.



Clustering Quality

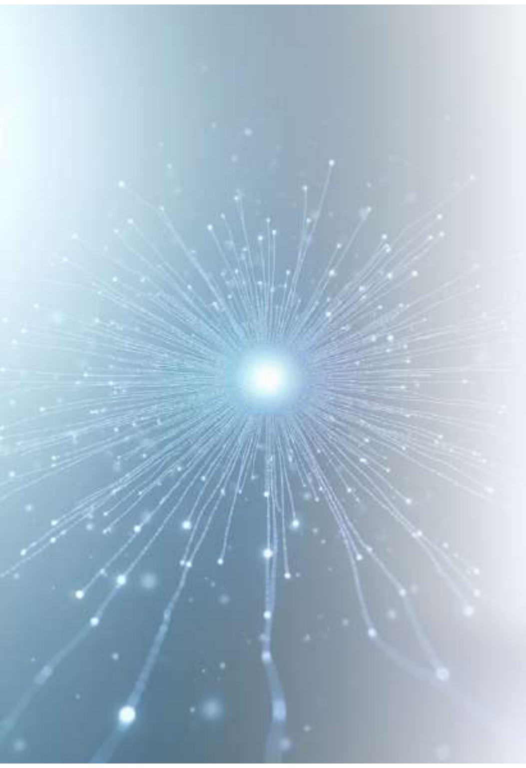
Higher value means better clustering.

Computation Speed

Fast to compute.

Cluster Shapes

Works well for spherical clusters.



Understanding Inertia in Clustering

What is Inertia?

Measures how close points are to their cluster center.

Lower Inertia Benefits

Lower inertia indicates tighter clusters.

Application in K-Means

Used mainly in K-Means.

Inertia and Cluster Count

Decreases as number of clusters increases.

Quick Summary of Metrics

Classification metrics

Measure label prediction quality

Regression metrics

Measure numeric prediction error

Unsupervised metrics

Measure cluster quality without labels